

Deep Learning

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WELCOME TO Deep Learning

- 1 References
- 2 What is Supervised Learning?
- 3 What is a neural network?

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville - Deep Learning
2. Original Papers
3. Blog Posts

Outline

- 1 References
- 2 What is Supervised Learning?
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What is Supervised Learning?

Definition 1

*In machine learning, **supervised learning** refers to a class of systems and algorithms that determine a predictive model using data points with known outcome.*

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- 1 Regression

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Supervised learning is often used to create machine learning models for two types of problems:

- 1 Regression
- 2 Classification

Example of regression problem

Example of regression problem

Size of House	Price of House
950	\$123,325
1,535	\$156,570
1,605	\$158,895
1,905	\$200,025
2,057	\$230,384
2,227	\$233,835
3,150	\$261,420
3,620	\$433,500

Example of classification problem

Example of classification problem



Our data consist of pairs

$$(x_i, y_i)_{i=1}^n, \text{ where } x_i \in \mathbb{R}^k, y_i \in \mathbb{R}^m, i = 0, 1, \dots, n.$$

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We have to find a function f for which

$$f(x_i) \approx y_i$$

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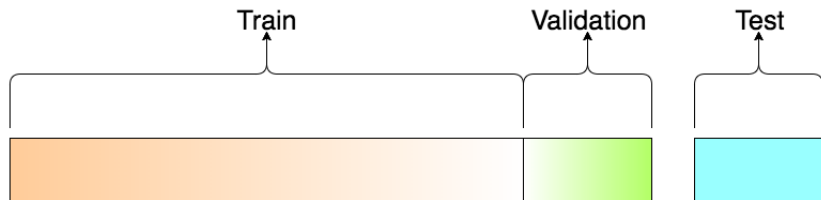
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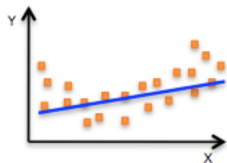
$$f(x_i) \approx y_i$$

and not only for pairs in our data.

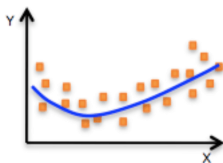
Train, Validation, Test



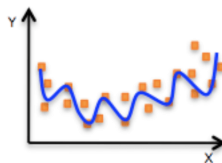
Overfitting, Underfitting



Underfitting

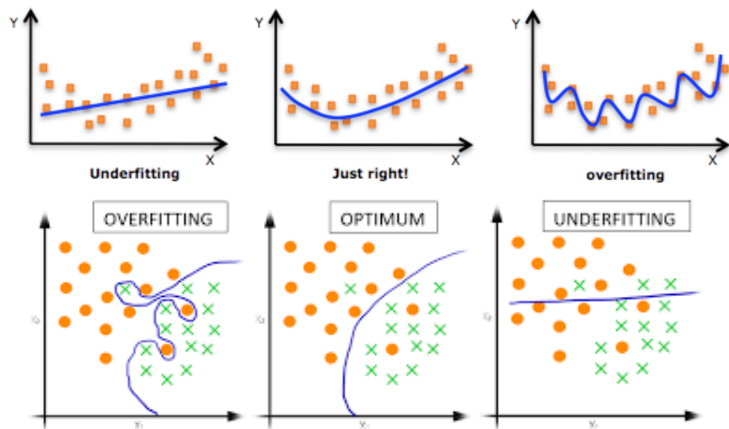


Just right!



overfitting

Overfitting, Underfitting



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Human Brain

